Neural Graphics
Assignment N.1

June 1, 2025

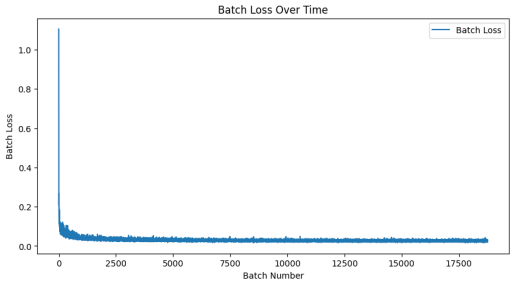
Bar, Alon
205476013

2 Unconditional Diffusion Framework

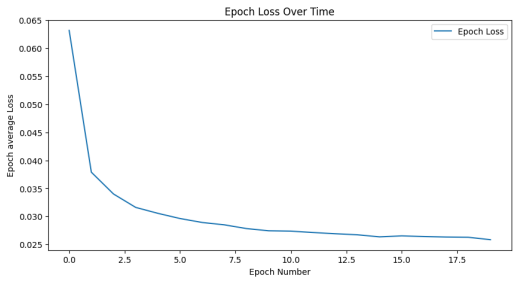
Different hyperparameters. I experimented with several modifications to improve performance. For the final convolutional layer, I tested kernel sizes of 3, 5, and 7, ultimately settling on a kernel size of 3 as it produced the best results. I also tuned the learning rate, trying both 1e−3 and 2e−4; the latter proved to be more effective. Additionally, I compared optimizers - while SGD was tested, Adam consistently outperformed it, as expected.

1. **Training loss curves**, showing both batch loss and epoch loss across the full training process. Please specify the axis labels in your plots.

Ans:



(a) Batch loss over steps



(b) Average loss over epochs

Figure 1: Training loss curves - showing both batch loss and epoch loss across the full training process of Unconditional Diffusion Framework

2. **Evaluation plots**, showing sampling results of the denoiser at epochs 1, 5, 10, 15 and 20. Each plot should contain eval batch size generated samples.

Ans:

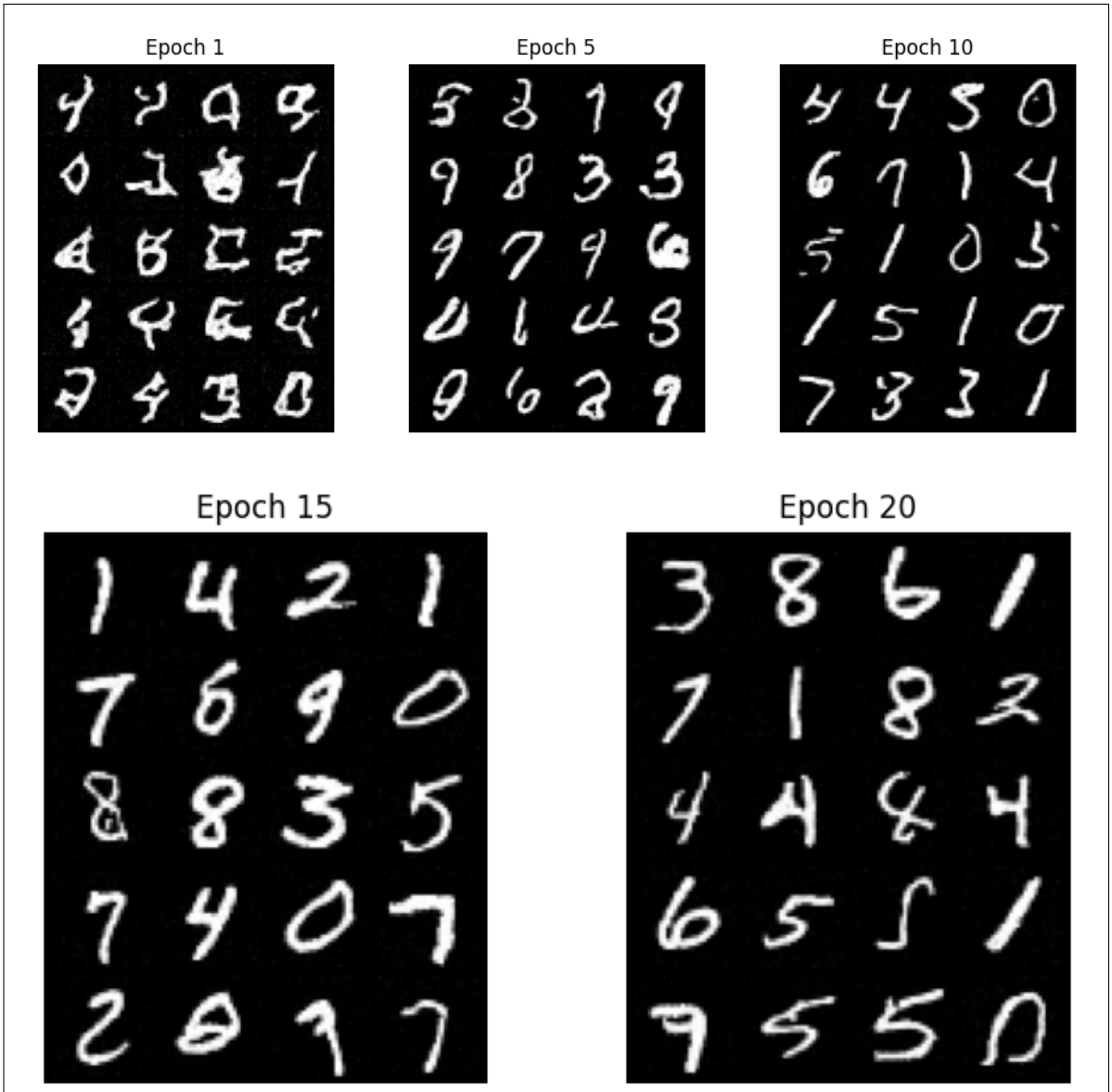


Figure 2: Evaluation plots, showing sampling results of the denoiser at epochs 1, 5, 10, 15 and 20 of Unconditional Diffusion Framework.

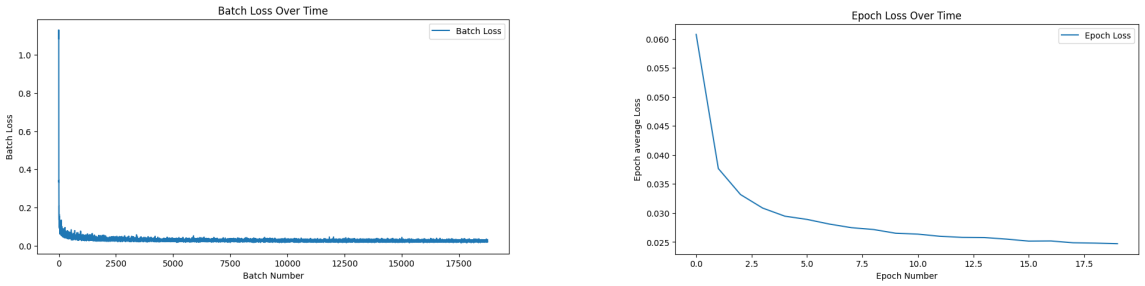
3 Implementing class-conditioned diffusion framework with CFG

Different hyperparameters. I experimented with several modifications to improve performance. Based on previous experiments, I used a kernel size of 3 for the final convolutional layer. I tuned the learning rate by testing both 1e−3 and 2e−4; in this case, the former yielded better results. I also compared optimizers — Again Adam was better.

1. **Training loss curves**, showing both batch loss and epoch loss across the full training

process. Please specify the axis labels in your plots.

Ans:



(a) Batch loss over steps

(b) Average loss over epochs

Figure 3: Training loss curves - showing both batch loss and epoch loss across the full training process of diffusion framework with CFG

2. **Evaluation plots**, showing sampling results of the denoiser at epochs 1, 5, 10, 15 and 20. For evaluation plots, use guidance scale $\gamma = 5.0$. Each plot should contain eval batch size generated samples of class-labels minibatches drawn from MNIST test set. Present each sample next to it's class-label.

Ans:

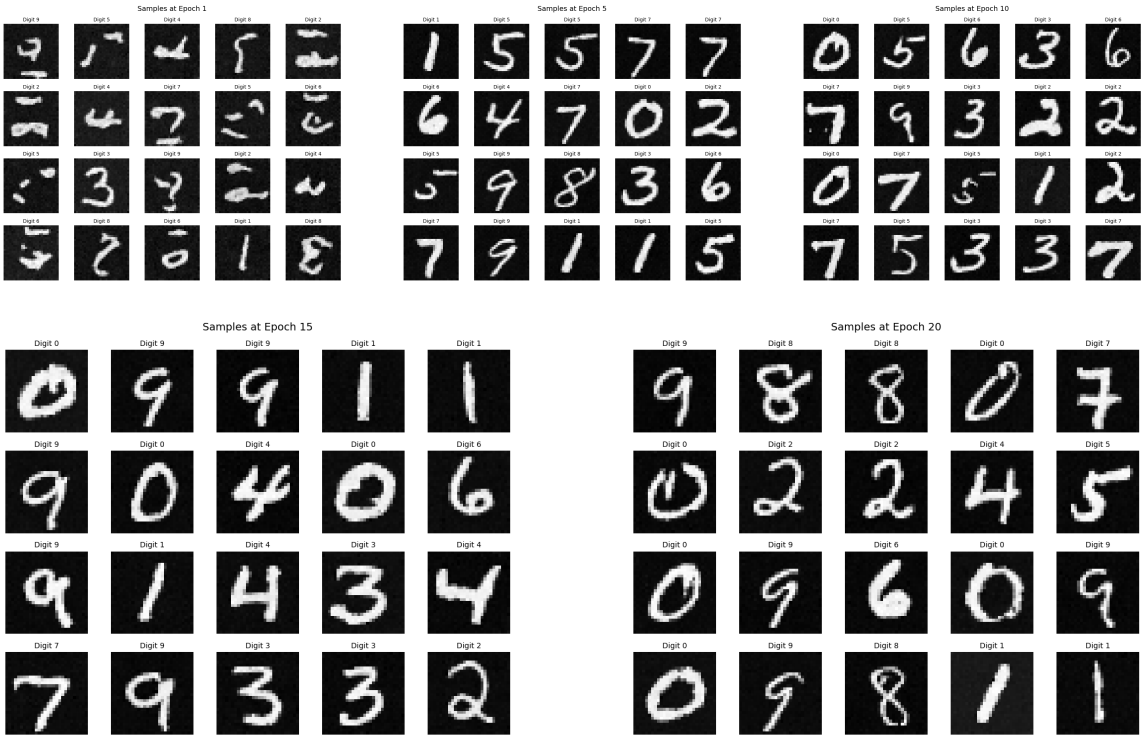
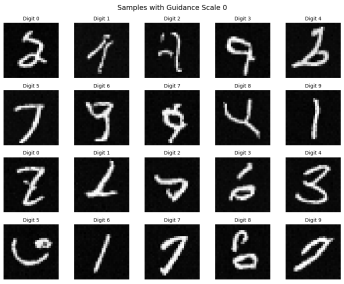


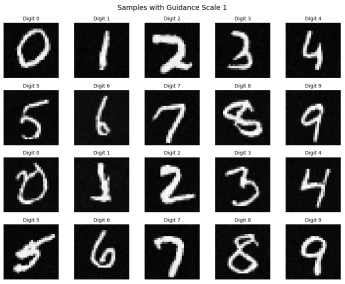
Figure 4: Evaluation plots, showing sampling results of the denoiser at epochs 1, 5, 10, 15 and 20 of diffusion framework with CFG

3. In this section, you will explore how varying the guidance scale impacts the generated images. Use your trained model to sample digits 0–9 with guidance scales of [0, 1, 5, 7, 10, 15]. Analyze and describe the visual effects of using guidance values that are too low or too high, and explain your observations based on the concepts discussed in class. Include both the generated images and your written analysis in your final submission.

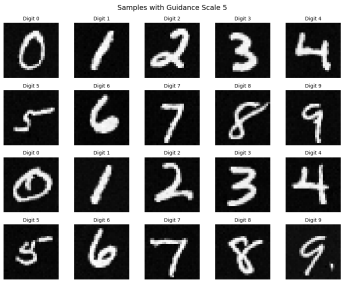
Ans:



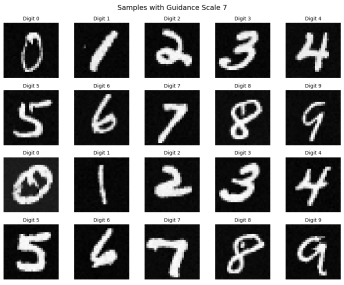
(a) Guidance scale - 0



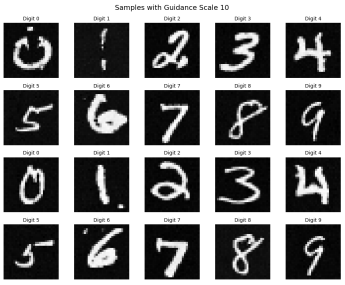
(b) Guidance scale - 1



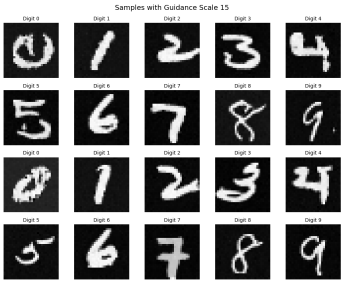
(c) Guidance scale - 5



(d) Guidance scale - 7



(e) Guidance scale - 10



(f) Guidance scale - 15

Figure 5: sample of digits 0–9 with guidance scales of [0, 1, 5, 7, 10, 15]

(a) **Guidance Scale = 0**
The generated digits are inconsistent and often fail to match the intended labels (e.g., “0” appears as “2”, “5” is noisy or malformed).
At $s = 0$, the model ignores the class condition entirely, resulting in purely unconditional samples that reflect the general data distribution without control.

(b) **Guidance Scale = 1**
Some class identity begins to emerge (e.g ”2” and ”7” ”8” and ”3”), but many digits are still ambiguous or fuzzy. Class-conditional accuracy is low.
With minimal guidance, the influence of the class label is weak. The samples lean toward being realistic but are not confidently aligned with the target digit.

(c) **Guidance Scale = 5**
All digits are clearly formed and correspond to the intended class. The class labels are perfectly matched. Visual quality and clarity are high.
The model achieves a good balance between conditioning on the class label and preserving sample diversity.

(d) **Guidance Scale = 7**
Digits become even sharper and more distinct. The class labels are perfectly matched.
Stronger guidance reinforces the conditional signal, producing high-confidence, clean digits. Slight loss of diversity may begin to appear (e.g the ”3”s are almost the same”)

(e) **Guidance Scale = 10**
Digits are mostly sharp and recognizable, but small artifacts and over-regularized patterns begin to appear — for example, digits like "0" and "1" look unnatural or overly stylized.
The high guidance scale pushes the model to rely heavily on the class condition. While this improves class consistency, it starts to introduce unnatural features due to reduced diversity and overstressed conditioning.

(f) **Guidance Scale = 15**
Some digits become distorted or unnatural (e.g., "0" and "5" appear with artifacts or exaggerated features).
Excessive guidance overpowers the model's learned noise prediction, pushing it too far toward the class signal, which destabilizes the generation and reduces realism.